

Common laboratory values

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Learning objective

- Explain the general principle of laboratory tests
- Identify the blood chemistry tests (Electrolytes, Cardiac Markers, Liver Function etc.)
- Identify the hematologic laboratory values (Hgb, WBC, Platelets, Erythrocyte sedimentation rate, Iron etc.)
- Differentiate abnormal lab values from normal (reference ones)
- Interpret different laboratory results of a patient

Introduction

Why laboratory testing?

- To detect diseases
- Guide treatment
- Monitor response to treatment
- Monitor disease progression
- Medico legal issues
- Genetic counseling in familial conditions

General principles

- Laboratory values must be assessed **in the context of the clinical situation.**
- They should not be evaluated in isolation of the subjective and objective findings.
- When interpreting laboratory test results, clinicians should use the normal values listed by **their own laboratory facility rather than those published in reference texts**, because laboratories may use different methods of assay.

General principles...

- Laboratory error should always be considered when laboratory results do not correlate with clinical expectations.
- Test reliability is impacted by various factors including statistical and preanalytical variations, accuracy, and precision.

General principles...

When to do laboratory tests?

- Laboratory tests should be ordered only if
 - The results of the test will affect decisions about the care of the patient.
 - The serum, urine, and other bodily fluids can be analyzed routinely;
 - The economic cost of obtaining these data must always be balanced by benefits to patient outcomes

Diagnostic Methods Classification

- **A. Biochemical Methods**

- Blood tests, Enzyme tests, Mineral content tests
- Test for evaluation of concentration - Na, K, Ca..., Glucose, Urea, in blood plasma

- **B. Physical Methods**

- Mechanical : e.g. Auscultation, Percussion, Palpation, Body temperature
- Electrical: ECG, EEG, Audiometry ...

Diagnostic Methods Classification...

- **Electrical Methods**

- **Electrocardiography (ECG)**-method for recording of heart electric signal from the surface of the skin.
- **Electroencephalography (EEG)**- a method for recording of brain electric signal from the scalp.
 - The point is evaluation of frequency (f) and amplitude(A) of waves e.g. in Epilepsy. Waves or Rhythms:

Diagnostic Methods Classification...

- **Ultrasound (Dopplers) methods:**
 - Blood flow test, Sonography, Echocardiography
- **X- ray imaging methods**
 - Classic Tomography, Computed tomography (CT)
- **Methods of Nuclear Medicine:**
 - Radio-isotopes , Gameography

Common Terms

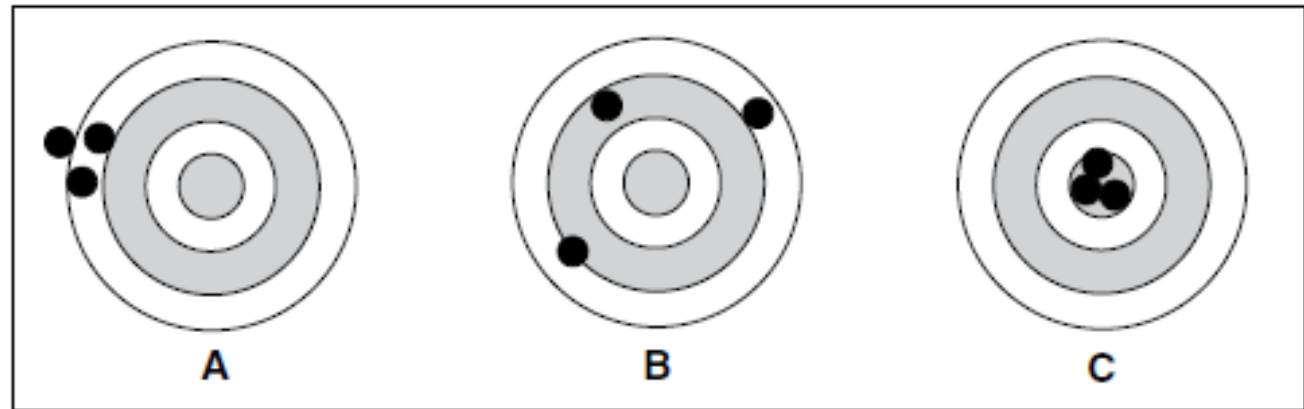
Precision

- Refers to the ability of the test to reproduce the same result
- when repeated on the same patient or sample
- The two concepts are related, but different.
- a test could be precise, but not accurate if on three occasions it
- produced roughly the same result, but that result differed
- greatly from the actual value determined by a reference standard

Common Terms ...

Accuracy

- The ability of the test to actually measure what it claims to measure,
- defined as the proportion of all test results (both positive & negative) that are correct



Common Terms ...

- **Sensitivity**

- Is the probability that a test is positive, given that the person has the disease

- **Specificity**

- Is the probability that a test is negative, given that the person does not have the disease

Basic Clinical Chemistry Normal Values

- Vary upon many factors
 - Age, gender, weight, height, race, comorbidity
- Normal values may vary from laboratory to laboratory , depending on techniques and reagents used.
- Each lab adapts its own methods and publishes its own normal value ranges
- Change in equipment or reagent

Serum Electrolyte Tests

Sodium (Na) (135 - 145 mEq/L)

- Sodium is the predominant cation of ECF.
- Only a small amount of sodium (~5 mEq/L) is in ICF.
- Along with **chloride, potassium, and water**, it maintains **serum osmolarity and osmotic pressure** relationships between ICF & ECF.
- Regulated by:
 - Aldosterone (enhances sodium reabsorption),
 - Natriuretic hormone (increases excretion of sodium), and
 - Antidiuretic hormone (enhances reabsorption of free water)

Sodium...

- Between 60-70% of the filtered load is reabsorbed in the proximal tubules.
- A further 25-30% is reabsorbed in the ascending loop of Henle.
- In adults, **changes in serum Na** concentrations most often represent **water imbalances rather than sodium imbalances**

Major Extracellular and Intracellular Electrolytes

ECF			ICF	
Ion Species	Plasma (mEq/L)	Interstitial Fluid (mEq/L)	Ion Species	mEq/L
Cations			Cations	
Na ⁺	142	144	K ⁺	135
K ⁺	4	4	Mg ²⁺	43
Ca ²⁺	5	2.5		
Mg ²⁺	3	1.5		
Total	154	152	Total	178
Anions			Anions	
Cl ⁻	103	114	PO ₄ ²⁻	90
HCO ₃ ⁻	27	30	Protein	70
PO ₄ ²⁻	2	2	SO ₄ ²⁻	18
SO ₄ ²⁻	1	1		
Organic acid	5	5		
Protein	16	0		
Total	154	152	Total	178

ECF, extracellular fluid; ICF, intracellular fluid.

Sodium...

- **Serum Sodium Abnormalities:**
 - Hyponatremia (<135 mEq/L)
 - Hypernatremia (>145 mEq/L)

Hyponatremia (Serum Na < 135 mEq/L)

- Decrease in the serum Na concentration could reflect.
 - Hypervolemia-**Dilutional Hyponatremia**
 - Abnormal Na losses-**Sodium-depletion Hyponatremia**
- Most remain asymptomatic until serum Na < 120 mEq/L

Hyponatremia...

Dilutional Hyponatremia

- Most cases of hyponatremia are caused by impaired water excretion in the presence of continued water intake.
 - Common in:
 - Cirrhosis, CHF, renal impairment,
 - Administration of osmotically active solutes (albumin ,manitol)

Sodium-depletion Hyponatremia

- Low serum Na concentration in the absence of edema.

Hyponatremia...

- **Antidiuretic Hormone (ADH)**

- Produced by hypothalamus (CNS)
- Also called vasopressin

- **Factors that may increase or decrease ADH production;**

- Hypovolemia, thirst, and high serum osmolality
- Leads to concentrated urine by increasing permeability of collecting tubules for re-absorption of water

Disorder of ADH Production...

- Syndrome of inappropriate ADH secretion (**SIADH**)
 - Inappropriately high quantity of ADH
 - Excessive reabsorption of water— **Hyponatremia**
- **Treatment**
 - water restriction
 - **Demeclocycline 600-1200 mg qd.**

Disorder of ADH Production...

- Diabetes Insipidus (DI)
 - DI (Central): Inadequate ADH synthesis
 - DI (Nephrogenic): kidney fails to respond to ADH– **hypernatremia**
 - **Drugs known to cause SIADH:**
 - Chlorpramide, Cyclophosphamide, Carbamazepine,
 - Tolbutamide, Oxytocin and some TCAs

Aldosterone

Is a mineral corticosteroid

- Retains Na and water – more Na than water
- Increase excretion of potassium
- Has more effect on K^+ than Na

Fractional Excretion of Sodium (1-2%)

A 24 hour urine sample should be used for estimation FENA.

$$FENa (\%) = \frac{\text{Urine Na/Serum Na}}{\text{UrineCr/Serum Cr}} \times 100$$

FENa > 2% refers to renal damage (more Na is excreted)

FENa < 1% refers to Na retention (e.g. heart failure)

Signs and symptoms

- **Mild:**

- may be asymptomatic;
- nausea,
- malaise,
- gait or cognitive disturbance

- **Moderate:**

- headache,
- lethargy,
- Restlessness and disorientation

- **Severe:**

- seizures
- coma
- respiratory arrest
- brainstem herniation
- death

Acute & Symptomatic Hyponatremia Treatment

- **Calculate the Na deficit**
- $\text{Na mEq} = ([\text{Na desired}] - [\text{Na measured}]) \times \text{TBW}$
- $\text{TBW} = 0.5(\text{female}) \text{ or } 0.6(\text{male}) \times \text{weight in KG}$
- Then replace **one-third of the deficit in the first 6 hours** and the remaining two-thirds over the following 24 to 48 hours
- Correct no faster than 1 mEq/L/hr for the first 6-8 mEq/L

Hyponatremia Treatment...

- Another aproch is:
 - Small amount of 3% NaCl
 - Then 0.9% NaCl can be used to continue the sodium correction.
 - No more than 10-12 mEq/L in first 24 hours

Hyponatremia...

Characteristics	Hypovolemic Hyponatremia	Euvolemic (Isovolemic) Hyponatremia	Hypervolemic Hyponatremia
Water and sodium	Sodium loss >> water loss	Water gain only	Water gain > sodium gain, heart failure
Causes	Renal: thiazide diuretics Nonrenal: diarrhea	Syndrome of inappropriate antidiuretic hormone (SIADH)	Heart failure Liver cirrhosis Kidney failure
Effect on TBW	↓↓	↑	↑↑
Effect on TBNa	↓	↔	↑↑
Laboratory findings in addition to hyponatremia	Renal: UOsm high, UNa high Nonrenal: UOsm high, UNa low	Renal: UOsm low, UNa variable Nonrenal: UOsm high, UNa variable	UOsm high, UNa high
Clinical presentation	Orthostasis, hypotension, tachycardia, dry mucous membranes, CNS changes	Depends on severity of hyponatremia: seizures, lethargy	Peripheral and pulmonary edema, variable blood pressure
Treatment	0.9% NaCl until vital signs stable; then maintenance fluid replacement (D5/0.45% NaCl)	Water restriction; demeclocycline; loop diuretics; vaptan	Sodium restriction; water restriction; loop diuretics; vaptan

Hypernatremia ($\text{Na} > 145 \text{ mEq/L}$)

- Less common – occurs in patients with impaired thirst mechanism (stroke patients)
- Serum $\text{Na} > 160 \text{ mEq/L}$ – 75% mortality
- Results from a deficit of water (**usually not due to excess Na intake**)
 - Loss of water (diarrhea, vomiting)
 - Failure to adequately replace the water loss
- Look thoroughly for **alterations in neurological status** that are causing inadequate water intake

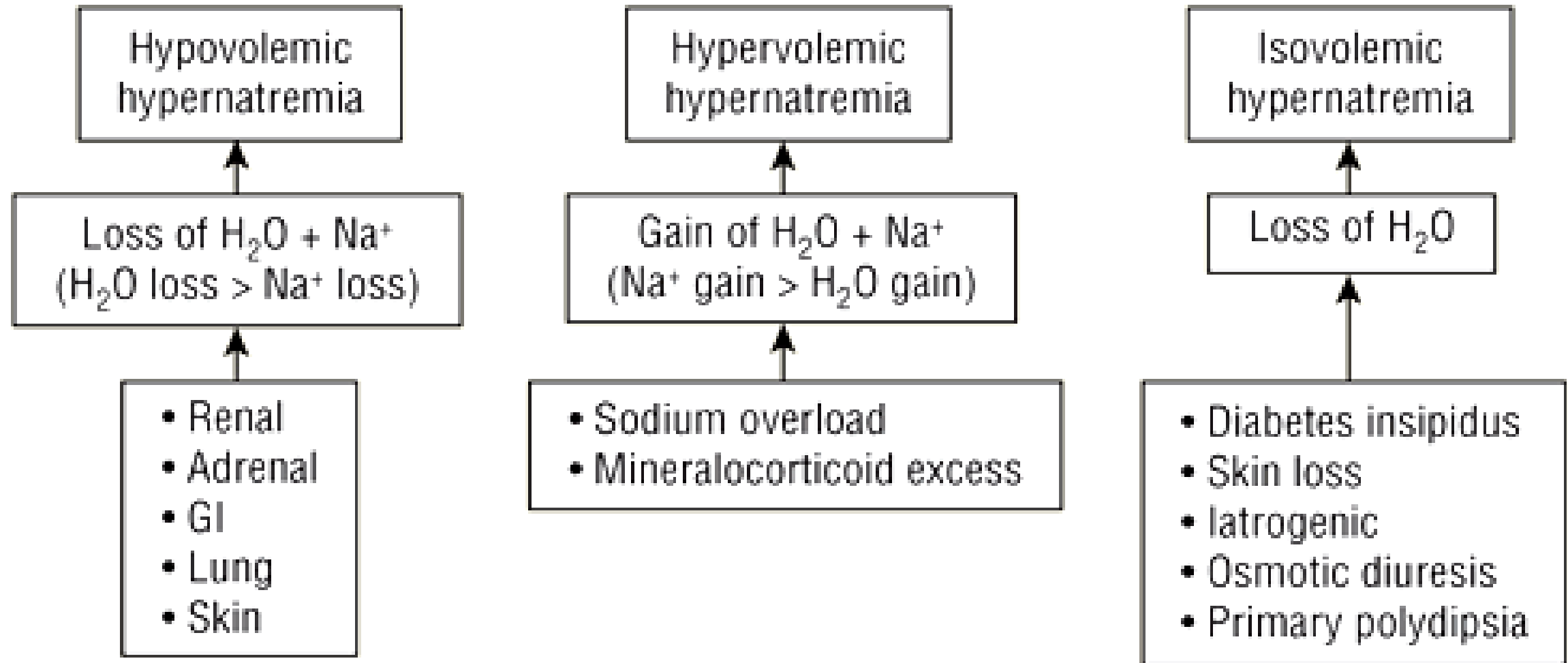
Hypernatremia

- Water loss is extra-renal or renal
- Extra-renal water loss
 - fever, profuse sweating, hyperventilation, diarrhea, vomiting
- Renal water loss (key is to look for polyuria)
- Key to evaluating renal water loss is to measure urine osmolality

Hypernatremia

- Diabetes Insipidus (urine Osm < 150)
- Inability of the kidney to concentrate urine due to absence of
- ADH (central) or
- unresponsiveness to ADH.

Hypernatremia



Signs and symptoms

Severe:

Mild:

- Lethargy
- weakness
- Confusion
- Restlessness
- Irritability

- Seizures
- coma
- Death
- usually requires acute increase in serum sodium to ≥ 160 mEq/L (mmol/L)

Treatment

- Calculate water deficit
- Severe **ECFV depletion is the priority and should be corrected** with NS first.
- Subsequent fluid replacement can be **hypotonic**
- Major complication of overly rapid correction is cerebral edema
- Safe rate is no more than 0.5- 1 mEq/L per hour
- Should take 36-72 hours to completely correct

Treatment

- Calculate the water deficit

$$H_2O \text{ deficit} = \frac{TBW \times ([Na \text{ measured}] - [Na \text{ desired}])}{[Na \text{ desired}]}$$

- Important to take into account ongoing losses
 - Insensible losses 0.5 - 1 liter/24 hours
 - With fever, these losses increase by 60-80ml/24 hrs for each degree Fahrenheit

Summary (Critical Values of Sodium)

- Hyponatremia: Less than 120 mEq/L
- Hypernatremia: Greater than 160 mEq/L
- Signs and symptoms
 - ▣ Hyponatremia
 - Confusion, irritability, convulsions, tachycardia, nausea, vomiting.
 - ▣ Hypernatremia
 - Restlessness, intense thirst, weakness, swollen tongue, **seizures and coma**

Potassium (K^+) (3.5 – 5.0 mEq/L)

- K^+ homeostasis is regulated through
- **Cellular K^+ shifts and K^+ excretion (renal or extra renal)**
- Disturbances in K^+ homeostasis result in **hypokalemia or hyperkalemia**
- **Hypo- and hyperkalemia**
 - Disrupt action potential formation and
 - promote various clinical symptoms and physical findings:
 - Neuromuscular dysfunction
- Rapid recognition and treatment of these K^+ disorders is required to avoid serious morbidity and mortality

Potassium...

- Total body K content is approximately 50 mEq per kg body weight. (K 3000–4000 mEq)
- Therefore, a 70 kg person has about 3500 mEq total body potassium
 - 98% of which is in the intracellular compartment.
- Serum K is associated with **Insulin, acid-base balance, renal function, GI losses**

Potassium...

- **Major role of Potassium :**
 - Regulate muscle and nerve excitability
 - Cardiac muscle depends on K for normal contraction

Potassium...

- A serum potassium concentration is not a good measure of total body K^+
- Fortunately, the clinical manifestations of K^+ deficiency correlate well with serum concentrations.
 - Fatigue, drowsiness, dizziness, confusion, electrocardiographic changes, muscle weakness, muscle pain)
- The serum K^+ concentration can be within normal limits despite abnormalities in total body potassium.

Potassium...

- **Potassium depletion**
- Moves from the ICF into the ECF to maintain the serum concentration.
- A mere decrease of serum K^+ by 0.3 mEq/L
 - leads to total body K^+ deficit ~ 100 mEq.

Hypokalemia (<3.5 mEq/L)

- The kidneys are responsible for about 90% of daily K^+ loss (~40–90 mEq/day),
- The remaining 10% is excreted by the GI system and the dermatologic system (sweating) each day .
- The kidneys have a limited ability to conserve K^+ .

Causes of Hypokalemia

- Inadequate oral intake
- Acid –base imbalance
 - **Metabolic alkalosis**
- Insulin, beta-2 agonists, and glucose cause shifting to Intracellular space.
- Renal impairment
- DKA
- Magnesium
- Aldosteronism
- Corticosteroid excess

Causes of Hypokalemia...

- Prolonged IV therapy with **K⁺-free solutions** in a patient unable to obtain potassium in foods (e.g., NPO] patient)
- Osmotic diuresis (e.g., mannitol, glucosuria),
- vomiting.
- Decreased food intake, loss of acid, alkalosis, and loss of sodium
- Skin **K⁺** loss
- Strenuous exercise
- Drugs

Complications (Clinical manifestations)

- **Neuromuscular manifestations**

- weakness, fatigue, paralysis, respiratory dysfunction

- **Gastrointestinal**

- constipation,

- **Cardiac**

- ECG changes
 - U waves, flattened T waves, arrhythmias

Hypokalemia-Management

- Mild: food (banana, orange, etc.)
- Moderate deficits (2.5–3 mEq/L),
 - Oral KCl (40–80 mEq/day) is preferred
- Severe: (<2.5)
 - IV KCl 20-40 mEq/L over 3-4 hrs
 - **Max infusion rate: 10 meq/h – faster rate leads to arrhythmia**
 - If ECG change with $[K] < 2 \text{ meq/l}$:
 - 40 meq/L/h-with frequent monitoring
 - Higher concentration leads to IV irritation/burning

Hypokalemia-Management...

- **Hypomagnesaemia** often accompanies hypokalaemia, as magnesium is necessary for the movement of **Na, K, and Ca** in and out of cells.
- Consequently, **hypokalemic** individuals will **not respond** to K therapy and may be refractory to treatment until **hypomagnesaemia** is corrected.

Hypokalemia-Management...

- Approximate the K^+ deficit as follows:
- The prevailing serum K^+ concentration
 - 3.0–3.5 mEq/L range, total body K^+ deficits reach 200–400 mEq
 - 2.0–3.0 mEq/L range, total body K^+ deficits reach 400–800 mEq

Hyperkalemia

- Hyperkalemia is defined as plasma K^+ concentration > 5 mEq/L.
- Pseudo hyperkalemia
 - Rarely falsely elevates the serum K^+ concentration
 - K^+ release from cells within the test tube
 - Cell lysis following prolonged tourniquet application
 - K^+ release from large cell numbers (white blood cells

True Hyperkalemia

- Impaired cell K^+ uptake (**intracellular shift**)
- Decreased renal K^+ excretion

Causes of True Hyperkalemia

- Dietary K^+ Excessive oral or IV intake (in combination with other factors)
- Cellular K^+ release/impaired cellular uptake
- Lack of insulin (fasting, diabetes mellitus)
- Metabolic acidosis
 - Decrease in pH by 0.1 from 7.4, it falsely elevate K^+ concentration by 0.6 mEq/L.

Causes of True Hyperkalemia

- **Drugs (ACEI, NSAID, Digoxin toxicity, eplerenone, β_2 adrenergic blockade)**
- **Renal K^+ retention**
 - Hypoaldosteronism
 - Hypoadrenalism (Addison's disease)
- Hyperkalemic, hyperchloremic metabolic acidosis

Clinical manifestations of hyperkalemia

- Incidence: 8% of hospitalized patients
- Severe **hyperkalemia is a medical emergency**
- **Respiratory failure** from diaphragmatic muscle weakness
- **Neuromuscular signs**
 - weakness, ascending paralysis, respiratory failure

Clinical manifestation...

- **Cardiac disturbances**

- Various AV nodal blocks,
- ventricular arrhythmias
- Abnormal myocardial contractile function causing hypotension
- **Progressive ECG changes**
- Peaked T waves, flattened P waves, prolonged PR interval, idioventricular rhythm and widened QRS complex

- Mortality could be as high as 67%

Treatment...

- Rapid therapy is required to prevent fatal outcomes
- Stabilization of excitable membranes (cardiac tissues if ECG Change)
- IV Ca^{2+} under cardiac monitoring;
 - Ca^{2+} should be given as a slower drip to patients on digoxin
- If no ECG change,
- Sodium polystyrene sulfonate in 50% sorbitol solution 15-30 gm qid prn.

Treatment...

- **Shift K^+ into cells**
- Intravenous regular insulin with glucose in nondiabetic patients
- β_2 adrenergic agonists
 - Lowers serum K^+ concentration (0.6–1.0 mEq/L)
- Combine insulin and β_2 adrenergic agonists
- Na^+ bicarbonate in patients with metabolic acidosis

Treatment...

- **Remove K^+ from the body**
 - Increase gastrointestinal K^+ excretion
 - Cation-exchange resin (**Na^+ polystyrene with sorbitol**)
 - Enhance renal K^+ excretion
 - High dose **loop diuretics**
 - **Hemodialysis** quickly removes K^+ from the body

Treatment...

- **Additional measures to treat hyperkalemia**
- Correct the primary cause of hyperkalemia
- Appropriately adjust dietary **K⁺** intake for level of kidney function

Chloride (95–105 mEq/L)

- Principal anion of the ECF and changes with Na concentration to maintain a neutral charge
- Serum [Cl] **has no real diagnostic significance** but it used to calculate the anion gap
- **Serves a passive role**
 - Maintenance of fluid balance and acid-base balance
- Useful in identifying fluid or acid-base balance disorders.
- **Increased Chloride:** seen in metabolic **acidosis**, **dehydration**, diabetes insipidus, and renal disorders

Chloride...

- **Decreased Chloride** [Hypochloremia]
 - shows metabolic alkalosis
 - May be associated with **prolonged vomiting**, gastric suctioning, metabolic alkalosis, CHF, SIADH, or use of acid suppressants.
 - May also result from significant diuresis.

Calcium (8.8 to 10.2 mg/dL)

- The total calcium content exist in **primarily in the bone**, with only about 1% freely exchangeable with that in the ECF.
- This **reservoir of calcium in bones maintains** the concentration of calcium in the plasma constant despite pronounced changes in the external balance of calcium.

Calcium...

- About 40% of the calcium in the ECF is **bound to plasma proteins** (especially albumin); 5% to 15% is **complexed with phosphate** and citrate; and about **45% to 55% is in the unbound, ionized form.**
- The serum total calcium $[Ca^{+2}]$ - Falls approximately 0.8 mg/dL for every 1 g/dL reduction in the **serum albumin** concentration

Calcium...

- Most laboratories measure the total calcium concentration, although it is the free, ionized calcium concentration that is important physiologically, and the form that is closely regulated physiologically.
- Most laboratories are also able to measure the ionized form of calcium (**4.5–5.6 mg/dL**)

Hypercalcaemia (>10.2 mg/dl)

- Hypercalcemia is the result of **one or a combination of three primary mechanisms:**
- Increased bone resorption, increased GI absorption, or increased tubular reabsorption by the kidneys.
- Increased flux of Ca^{2+} into the ECF from
 1. Skeleton
 2. Kidney
 3. Intestine

Hypercalcaemia...

❖ Severity

❖ Mild or Moderate

❖ **Ca²⁺** -10.2- 13.0 mg/dL

❖ Severe

❖ **Ca²⁺** > 13.0 mg/ dL

Hypercalcaemia...

❖ Symptoms

- Weakness, fatigue, anorexia, depression, anxiety and cognitive dysfunction
- **GI:** Constipation, abdominal pain, anorexia, NV, peptic ulcer
- **Renal:** Polyuria, polydipsia, and nocturia

❖ Signs

- **Cardiovascular:** ventricular tachyarrhythmia's and ↑BP.
- **Musculoskeletal:** Gout, pseudo-gout, chondrocalcinosis/cartilage
- **CNS:** Drowsiness, lethargy, altered mental status, coma

Hypercalcaemia: Treatment

- ❖ In CKD stages 3 & 4, initial treatment of severe hypercalcaemia:
 - ❖ Infusion of normal saline 200 to 300 mL/h
 - ❖ **Continue until** the patient is fluid resuscitated and serum calcium approaches the ULN
 - ❖ **Add Lasix**
 - ❖ 40–80 mg IV every 1–4 hours or ethacrynic acid
 - ❖ **Add Potassium chloride**
 - ❖ 10 to 20 mEq/L to the saline solution after rehydration is accomplished to maintain **normokalemia** in the presence of **diuretic therapy**

Hypocalcaemia

- **Implies** a deficiency in either the **production of or response to parathyroid hormone or vitamin D**

❖ Causes

- Hypoalbuminemia
- Chronic renal failure
- Magnesium deficiency
- Hypoparathyroidism
- Vitamin D deficiency (or resistance)
- Drugs: Phenytoin, Phenobarbital, and Rifampin

Hypocalcaemia: causes

❖ Calcitonin

❖ Is a hormone secreted by Thyroid gland.

- In response to increase in Ca (ionized)
- **Inhibits osteoclastic activity** thus **decrease bone resorption**

❖ Increase in renal calcium clearance

○ Loop diuretics

❖ **Phosphate** binds Ca leads to decrease Ca (by forming insoluble product)

Hypocalcaemia Signs and symptom

- ❖ **Neurologic:** Tetany, mainly parkinsonism,
- ❖ **Dermatologic:** Dry skin, hyperpigmentation, dermatitis
- ❖ **Hair and skin signs:** Coarse, brittle, and sparse hair with patchy alopecia
- Ophthalmologic:** Cataract
- ❖ **Dental manifestations:** Dental hypoplasia, failure of tooth eruption, defective enamel and root formation

Signs and symptom...

- ❖ **Cardiovascular:** Hypotension, decreased myocardial performance, and CHF, prolonged QT interval, arrhythmias, and bradycardia
- ❖ **Gastrointestinal:** Steatorrhea (fatty stool)
- ❖ **Musculoskeletal:** Skeletal abnormalities like myopathy
- ❖ **Endocrine:** impair insulin release

Hypocalcaemia: Treatment

❖ Acute symptomatic

- About 1 to 2 g of calcium gluconate in 50 mL of 5%DNS infused **over 10 to 20 minutes.**
- If Infusion rate **> 60 mg Ca/min.**
 - Risk of severe **cardiac dysfunction**, including **ventricular fibrillation.**
- After correction of acute cause by IV admin, individualize further treatment be based on **cause of hypocalcemia**

Magnesium

- The normal range for serum magnesium is 1.4 to 1.8 mEq/L
- Primarily an intracellular electrolyte that, together **with K and Ca**, helps **maintain a neutral charge within the cell.**
- About 55% present in skeleton
- About 1% of total body Mg extracellular

Magnesium...

- Required for ATP
- Glycolysis
 - Cell replication
 - Protein biosynthesis

Hypermagnesaemia

❖ Symptoms

- Depressed neuromuscular system
- Depressed respiration
- Cardiac arrest

❖ Causes

- Excessive intake
- Antacids -Moxal
- Enemas
- Mg administration (in patients with RF)

Hypomagnesaemia

- Common in patients with multiple electrolyte disturbances
- Usually associated with **hypokalaemia and hypophosphatemia**
- Increased neuromuscular excitability

Hypomagnesaemia

- **The three most common causes:**
 - Diabetes
 - Alcoholism
 - Use of diuretics
- **Other Drugs**
 - Aminoglycosides, amphotericin B, Cyclosporine, Osmotic diuresis (DM/mannitol) and pentamidine
- **Renal loss**
 - Chronic TPN, Hypercalcaemia, Renal disease
- **Gastrointestinal**
 - Prolonged nasogastric suction, malabsorption, Bowel resection, Diarrhea, Acute pancreatitis, Chronic vomiting

Hypomagnesaemia treatment

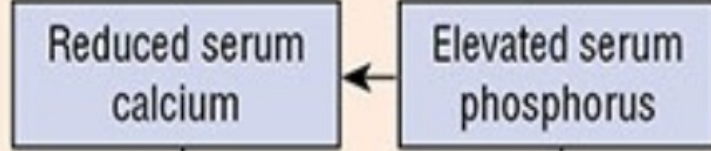
- Replacement iv for level < 1 mg/dL (0.32 mmol/L)
 - Emergent administration over 5–10 mins
 - Less urgent administration over 10–60 mins

Phosphate Metabolism (2.6 – 4.5 mg/dL)

- A major intracellular anion and is the source of ATP and phospholipid synthesis.
- 85% present in skeleton 10% protein bound, 35% complexed, rest free
- Serum calcium and PO₄ are influenced by many of the same factors.
- Consider together with Ca when interpreting lab results

Phosphate...

- Its ECF concentration is **influenced by**
 - PTH
 - intestinal phosphate absorption
 - renal function
 - Nutrition
 - bone metabolism



Increased parathyroid hormone secretion

1. Increased osteoclast and osteoblast activity in bone
Increased calcium mobilization

2. Increased renal calcium reabsorption
Decreased renal phosphorus reabsorption

3. Increased renal activation of 1,25-dihydroxy vitamin D₃

Increased serum calcium

An increase in calcitonin

Increased intestinal calcium and phosphorus absorption

Hypophosphatemia

- Occur in **malnutrition**, in **chronic alcoholics**, and in **septic patients**.
- **Hyperparathyroidism** and **insufficient vitamin D intake** can also induce hypophosphatemia.
- Clinical consequences of severe hypophosphatemia may involve:
 - Muscle weakness
 - Cardiac irregularities, and
 - Dysfunction of leukocytes and erythrocytes
- **Treatment**
 - Acute hypophosphatemia usually requires IV phosphorus supplementation.

Hyperphosphatemia...

- **Caused by**
 - Renal insufficiency
 - Increased vitamin D
 - Increase phosphate intake
 - Hypoparathyroidism, and
 - Hyperthyroidism are also causes

Thank you